MIL-T-45540B
31 August 1973
SUPERSEDING
MIL-T-45540A
30 September 1965

MILITARY SPECIFICATION

TARGET HOLDING MECHANISMS, TRAINFIRE, M30 AND M30A1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of electro-mechanical target mechanisms, with remote control equipment (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-915/6	-	Cable, Electrical, Special Purpose, 600 Volts, A.C., Types SHOF, DHOF, THOF, and FHOF.
MIL-C-915/14	-	Cable, Electrical, Special Purpose, 600 Volts, A.C., Type MHOF.
MIL-C-13777/1	-	Cable, Special Purpose, Electrical 2, 3, 4 and 5 Conductors.
MIL-P-14232	-	Parts, Equipment and Tools for Army Material, Packaging and Packing of.

STANDARDS

Federal

FED-STD-595	-	Colors

FSC 6920

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Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-171	-	Finishing of Metal and Wood Surfaces.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.

DRAWINGS

Army - WC

F7546151	-	Target Holding Mechanism, M30.
F7546152	-	Wiring Diagram for M30.
F8448232	-	Target Holding Mechanism, M30A1.
F8448233	-	Wiring Diagram for M30A1.
F12002898	-	Target, Silhouette, Type "E" Kneeling, Polyethylene.

PUBLICATIONS

Army - WC

P7546151	-	Packaging Data Sheet, Target Holding Mechanism, Trainfire, M30.
P8448232	-	Packaging Data Sheet, Target Holding Mechanism, Trainfire, M30A1.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposed shall apply.

American National Standard Institute (ANSI)

ANSI-B46.1	-	Surface Texture.
ANSI-Y14.5	-	Dimensioning and Tolerancing for Engineering Drawings.
ANSI-Y32.3	-	Welding Symbols.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY, 10018)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Material. Material shall conform to the material specifications referenced on the detail drawings of Drawing F7546151 for the M30 Target Mechanism and F8448232 for the M30A1 Target Mechanism.

3.2 Dimensions. Dimensions of parts and assemblies shall conform to the applicable drawings of F7546151 for the M30 and F8448232 for the M30A1. Interpretation of dimensional requirements shall be in accordance with ANSI-Y14.5.

3.3 Construction.

3.3.1 Assembly. Parts shall be positioned and attached in accordance with the applicable drawings of F7546151 for the M30 and F8448232 for the M30A1.

3.3.2 Welding. Spot-welding and arc-welding shall be in accordance with the specifications referenced on the applicable detail drawings of F7546151 for the M30 and F8448232 for the M30A1. Interpretation of weld symbols shall be as defined in ANSI-Y32.3. Actual Production of weldments shall not proceed prior to Government approval of the contractor's welding procedures and test specimens.

3.3.3 Soldering. The soldering of electrical connections shall be in accordance with requirement #5 of MIL-STD-454.

3.4 Protective finishes.

3.4.1 Inorganic finishes. Inorganic finishes shall be in accordance with the finish number of MIL-STD-171 specified on the detail drawings of F7546151 for the M30 and F8448232 for the M30A1.

3.4.2 Organic finishes. The target mechanisms and parts to be painted shall be in accordance with the finish number of MIL-STD-171 specified on the detail drawings of F7546151 for the M30 and F8448232 for the M30A1. Parts not to be painted shall be as specified in the same standard. Unless otherwise specified in the contract or order, the finish number specified in MIL-STD-171 for cleaning shall be optional with the contractor.

3.4.3 Colors. Colors as specified on drawings shall be 27875 White, 21005 Red, 27038 Black and 24087 Olive Drab of FED-STD-595.

3.5 Surface roughness. Interpretation of surface roughness symbols shall be as defined in ANSI-B46.1.

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3.6 Performance and product characteristics.

3.6.1 Power input. Each target mechanism shall operate as specified herein on a single phase alternating current input of 115 volts, plus or minus 10 percent, and a frequency of 60 cycles, plus or minus 5 percent. Current consumption shall not be more than 4 amperes.

3.6.2 Operation. Each target mechanism shall perform as follows:

3.6.2.1 Manual operation. Each mechanism shall raise and lower dual Targets, F12002898, when controlled by the operator. The time required to raise the dual targets shall not exceed 3.0 seconds, at a temperature of 75° + 10° Fahrenheit. The time required to lower the same dual targets shall not exceed 2.0 seconds, at a temperature of 75° ± 10°F.

3.6.2.2 Automatic operation. Each mechanism shall register individual hits and shall lower (kill) dual Targets, F12002898, automatically when hit with automatic and semi-automatic small arms fire of Cal. .30 or 7.62MM ammunition, at a distance of 25 yards to 800 yards in an outdoor area.

3.6.2.3 Recording device. Each target mechanism shall be equipped with a recording device of a four digit, manual reset type which will register each hit on the targets.

3.6.3 Water-tightness. The Control Box Assembly (Part No. 7546146) and the Junction Box Assembly (Part No. 7546148) pertaining to the M39 Target Mechanism and the Mechanism Box Assembly (Part No. 8448234) and the Hit-Switch Assembly (Part No. 7546054) pertaining to both target mechanisms shall not leak when the target holding mechanism is submerged with three feet of water above the highest point of the equipment, for a period of five minutes. The temperature of the water shall be 75° ± 10°F.

3.6.4 Reliability. Each target mechanism with dual Targets, F12002898, in place shall operate for a minimum of 240 cycles at the rate of 3 cycles per minute, at zero degree, ± 5°F. A ten cycle warm-up at zero degree temperature, ± 5°F., is permitted. Each target mechanism with the dual targets in place shall operate for a minimum of 240 cycles at the rate of 4 cycles per minute, at 125 degrees ± 5°F. Each target mechanism with the dual targets in place shall operate continuously at 75 degrees ± 10°F., for a minimum of 9600 cycles at 4 cycles per minute.

3.7 Marking. Identification marking of the target holding mechanism and applicable parts shall be in accordance with the drawings and MIL-STD-130. Serial numbers shall be furnished by the procuring activity (see 6.3).

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3.8 Government-furnished property. Targets conforming to Drawing F12002898 shall be furnished the contractor for test purposes. Quantity of targets shall be as specified by the procuring activity (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplier and services conform to prescribed requirements.

4.2 Preliminary inspections. Prior to initiating acceptance inspection, two completed identical target holding mechanisms, taken at random from the initial lot, shall be subjected to the automatic operation test of 4.5.3.2 and the reliability test of 4.5.4. If either of the two mechanisms fail to pass these tests, acceptance inspection shall be deferred pending corrective action by the contractor. After Government verification of corrective action by the contractor, the test shall be repeated with two similar target holding mechanisms other than the two originally tested. Acceptance inspection shall be initiated after two mechanisms have passed this test. The disposition of the mechanisms used in this test shall be determined by the contracting officer. Unless otherwise specified in the contract or order, the tests of 4.5.3.2 and 4.5.4 shall be repeated on one target holding mechanism taken at random from each 1000 mechanisms produced.

4.3 Inspection provisions.

4.3.1 Lot formation. Unless otherwise specified in the contract or order, a lot shall consist of all items manufactured or packaged under essentially the same conditions.

4.3.2 Sampling for examination. A random sample of target holding mechanisms shall be taken from each lot in accordance with MIL-STD-105, at Inspection Level II. Acceptable Quality Levels (AQL) shall be as specified in Table I.

4.3.2.1 Sampling for packaging and packing. Sampling for examination of packaging, packing, and marking shall be in accordance with MIL-P-14232.

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4.3.3 Sampling for water-tightness test. A random sample of target holding mechanisms shall be taken from each lot in accordance with MIL-STD-105, at Inspection Level S-4. The Acceptable Quality Level shall be 1.5 percent defective.

4.3.4 Certification.

4.3.4.1 Material. The contractor shall furnish certification that materials used in construction of the target holding mechanism conform to the specifications referenced on the applicable drawings.

4.3.4.2 Protective finishes. The contractor shall furnish certification that cleaning and inorganic and organic finishes (painting) conform to the requirements of the applicable drawings and referenced specifications and standards.

4.4 Examination.

4.4.1 Each of the target holding mechanisms taken in accordance with 4.3.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in Table I.

Table I - Classifications of Defects

Categories	Defects	Method of Inspection
Critical	None defined	
Major	AQL = 1.0 percent defective	
101	Mechanism incomplete, parts missing	Visual
102	Missing welds, or visible defects of welding	Visual
103	Visible soldering defects	Visual
Minor	AQL = 4.0 percent defective	
201	Painting, including runs, blisters, sags, incomplete coverage and wrong color	Visual
202	Identification marking and instructions missing, incomplete or illegible	Visual

4.4.2 Packaging, packing and marking. Examination of packaging, packing and marking shall be performed in accordance with MIL-P-14232.

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4.5 Test procedures.

4.5.1 Equipment. Each target holding mechanism taken for test shall be equipped with the following parts:

a. Two (2) Targets (silhouettes), F12002898.
b. One cable conforming to cable Type DHOF-9, of MIL-C-915/6.
c. One cable conforming to cable Type MHOF, of MIL-C-915/14.
d. One cable conforming to cable Type 030565 of MIL-C-13777/1.
e. The targets shall be installed on the target holder assemblies.
f. The cables, cut to suitable lengths for conducting the tests, shall be installed in accordance with Wiring Diagram F7546152 for the M30 or F8448233 for the M30A1 and then connected to the power source specified in 3.6.1.

4.5.2 Water-tightness. Each target holding mechanism taken in accordance with 4.3.3 shall be tested for compliance with 3.6.3. Upon completion of the test, the presence of water in any box assembly shall be considered defective. Caution shall be taken during tests and subsequent inspection to avoid confusion of condensation with leakage.

4.5.3 Operation. Target holding mechanisms assembled with the equipment of 4.5.1, and connected to the power source specified in 3.6.1, shall be tested as follows:

4.5.3.1 Manual operation. Each target holding mechanism produced shall be placed in an area having a temperature of 75 degrees ± 10° Fahrenheit. The dual targets shall be lowered manually by tapping with the finger and raised by operating the control switch. The time required to raise or lower the targets shall be measured with an appropriate timing device to determine conformance to 3.6.2.1. Any mechanism failing this test shall be rejected.

4.5.3.2 Automatic operation. Target holding mechanisms taken in accordance with 4.2 shall be placed in an outdoor area with the dual targets in the "up position." With the two targets in place, 10 rounds of semi-automatic small arms fire and three 5 round bursts of automatic small arms fire of either Cal. .30 or 7.62MM ball ammunition shall hit each target from a distance of 50 ± 5 yards to determine conformance to the requirements specified in 3.6.2.2.

4.5.4 Reliability. Target holding mechanisms taken in accordance with 4.2 shall be equipped as specified in 4.5.1 and tested as follows:

a. The mechanism shall be subjected to zero degree ± 5°F. temperature for a minimum of 6 hours and then functioned 240 cycles at the

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rate of 3 cycles per minute, while still in the cold room. A 10 cycle warm-up at the zero degree ± 5°F. temperature is permitted.

b. The mechanism shall then be subjected to 125 degrees ± 5°F. temperature for a minimum of 6 hours and then functioned 240 cycles at the rate of 4 cycles per minute, while still in the hot room.

c. After successfully passing the above tests, the mechanism shall be operated continuously for 9600 cycles at a temperature of 75 degrees ± 10°F. at 4 cycles per minute.

d. The movement of the targets from the vertical position to the horizontal position and return to the vertical position shall constitute one cycle.

4.5.5 Recording device. During the test of 4.5.4c, the recording device shall be examined to determine compliance with 3.6.2.3.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking of the Target Holding Mechanism shall be in accordance with Packaging Data Sheet P7546151 for the M30 and P8448232 for the M30A1.

5.2 Levels. The level of protection required shall be as specified in the contract or order (see 6.2).

6. NOTES

6.1 Intended use. The Target Holding Mechanisms are designed as remote scoring targets for use in markmanship training in small arms firing. Each mechanism, with appropriate silhouette targets, may also be used for identification training for recognition of machine gun nests, squads of men, etc.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Model number.
c. Title, number and date of applicable drawings.
d. Place of inspection and test.
e. Level of protection required (see 5.2).
f. Quantity of targets to be furnished the contractor (see 3.8).
g. Disposition of reliability tested mechanisms (see 4.5.4).
h. Responsibility for test firing facilities and operating procedures (see 4.5.3.2).
i. Responsibility for furnishing ammunition (see 4.5.3.2).

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6.3 Serial numbers. The contractor should be furnished a block of serial numbers by the contracting officer, to be used for marking the target holding mechanism nameplates.

6.4 Supersession data. This specification includes the requirements for the following NAVTRADEVCEN specifications.

1. 3131-236 - Target Set, Disappearing "E" or "F" Pop-up 3C52E.
2. 3131-279 - Target Set, Silhouette, Disappearing "E" or "F" Pop-up, 3C52B (Marine Corps).
3. 3131-284 - Target Set, Silhouette, Disappearing "E" or "F" Pop-up, 3C52B (Army).
4. 3131-307 - Target Set, Dual Silhouettes, Disappearing "E" or "F" Pop-up 3C52B-2.
5. 353-203 - Target Set, Silhouette, Remote Controlled Device 3C52C, supersedes 3131-307.

Custodians:
Army - WC
Navy - TD
Air Force - 70

Review activities:
Army - None
Navy - None
Air Force - None

User activities:
Army - None
Navy - MC
Air Force - None

Preparing activity:
Army - WC

Project No. 6920-0065

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-T-45540B - Target Holding Mechanisms, Trainfire, M30 and M30A1.

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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